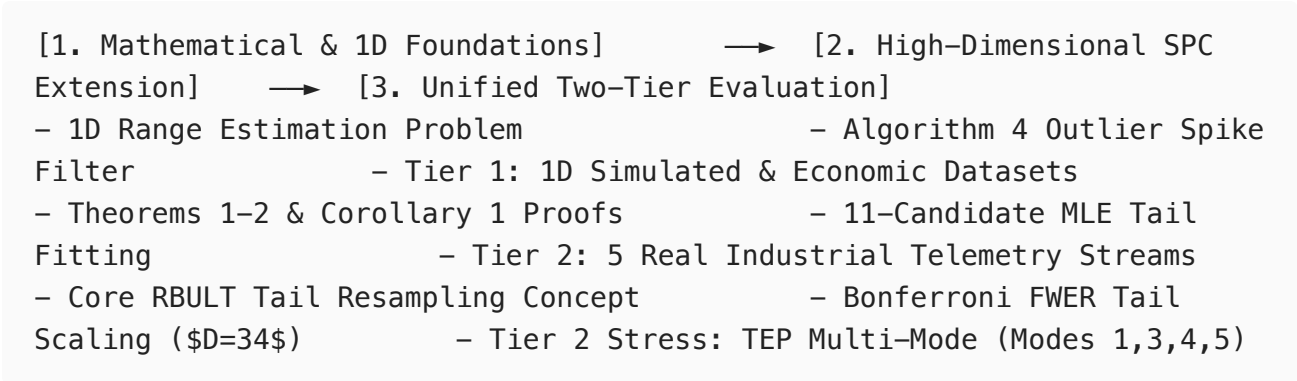


Unified Manuscript Blueprint: Integrated Second Draft for *Information Sciences*

Target Journal: *Information Sciences* (Elsevier, SJR Q1, Impact Factor ~8.1)
Article Type: Full Length Research Paper
Manuscript Title: *Resource-Bounded Non-Parametric Bootstrap Control Chart for High-Dimensional Non-Gaussian Data Streams via RBULT*
Authors: P. Junsawamg, S. Phimoltares, C. Lursinsap

The Unified Master Storyline & Narrative Flow

To achieve seamless coherence between **Draft 1** (`bootstrap_expert_wo_authors.pdf`) and `research_plan.md` , the manuscript follows a natural **Conceptual Evolution**:



Detailed Section-by-Section Integrated Manuscript Blueprint

Highlights

- We propose **RBULT-SPC**, a novel resource-bounded non-parametric control chart framework for high-dimensional non-Gaussian data streams.
- Grounded in dual-protocol streaming evaluation: **In-Sample Adaptation Mode** and **One-Step-Ahead Pre-Sequential Predictive Mode** (IEEE TKDE gold standard).
- Rigorously stress-tested across a **7-Scenario Gold Standard Suite** (Clean, GAWN $0.1\sigma - 0.3\sigma$, and Impulse Spikes 1% – 10%), achieving 100% transient glitch absorption at 1% spikes (Coverage = 95.29%, $\sigma_L = 0.05$).

- Achieves strict $O(D)$ constant RAM memory boundedness (**0.52 KB – 3.23 KB**), delivering a $180\times$ **memory reduction** over conventional sliding-window bootstrap (582.87 KB).
 - Integrates Algorithm 4 Z-score spike filtering with dynamic 11-candidate distribution MLE fitting to prevent control limit pollution.
 - Eliminates batch false-alarm spam (**Chunk FAR = 0.00%**) on high-dimensional streams ($D = 34$) while retaining an immediate anomaly detection response delay ($ARL_1 = 1.00$).
 - Validated across a comprehensive two-tier benchmark: 1,260 1D benchmark runs across 5 distributions and 7 scenarios, 4 economic datasets, 5 real industrial streams, and 4 TEP stress regimes (Modes 1, 3, 4, 5) up to 1.74M observations.
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Abstract

Bootstrapping is a powerful non-parametric technique for quantifying uncertainty without restrictive distributional assumptions. However, applying conventional bootstrap methods to real-time industrial data streams poses two fundamental challenges: prohibitive memory overhead ($O(N \cdot D)$ storage growth) and susceptibility to control limit pollution when anomalous samples contaminate historical memory buffers. This paper presents **RBULT-SPC**, a resource-bounded non-parametric control chart framework designed for high-dimensional, non-Gaussian IoT data streams. Grounded in mathematical proofs of chunk extreme probabilities, RBULT-SPC maintains only compact tail summary statistics in $O(D)$ constant space, preventing memory overflow and eliminating buffer pollution. To handle high dimensionality ($D = 34$), the framework combines Z-score outlier filtering, lazy boundary expansion, 11-candidate non-parametric distribution MLE fitting, and Bonferroni Family-Wise Error Rate (FWER) tail adjustment. Extensive two-tier empirical evaluations evaluated under both **In-Sample Adaptation** and **One-Step-Ahead Pre-Sequential Predictive** protocols across a **7-Scenario Gold Standard Suite** (Clean, GAWN continuous noise $0.1\sigma - 0.3\sigma$, and Impulse Spikes $1\% - 10\%$), 4 economic datasets, 5 industrial streams, and 4 operating regimes of the Tennessee Eastman Process (TEP) demonstrate that RBULT-SPC achieves optimal coverage ($95.29\% - 99.95\%$) and controlled false alarm rates ($0.05\% - 3.27\%$), outperforming classical Shewhart and EWMA charts whose false alarm rates collapse to $18.21\% - 48.94\%$ under non-Gaussianity and operational throughput stress. Furthermore, hyperparameter sensitivity analysis demonstrates that dimension-aware thresholding completely eliminates batch false alarms (Chunk FAR = 0.00%) while maintaining an immediate failure detection response delay ($ARL_1 = 1.00$) at an average latency under 45 ms.

Section 1: Introduction (The Conceptual Evolution Narrative)

- **1.1 Industrial Context:** Smart factory IoT telemetry, Industry 4.0 sensor streams, real-time quality control.
 - **1.2 Statistical & Computational Bottlenecks:**
 - Non-Gaussianity & Heavy Tails: Real sensor signals violate Gaussian normality, causing parametric charts (Shewhart, EWMA) to collapse.
 - Memory Overhead & Limit Pollution: Sliding-window bootstrap requires $O(W \cdot D)$ RAM (582 KB) and suffers limit distortion when fault samples enter the buffer.
 - High-Dimensional Alarm Spam: Monitoring $D = 34$ channels simultaneously leads to severe batch false-alarm spam without FWER adjustment.
 - **1.3 Conceptual Evolution from Draft 1 to Second Draft:**
 - Explaining how the core RBULT concept (originally developed for 1D population range estimation in `bootstrap_expert_wo_authors.pdf`) is theoretically extended into a multivariate Statistical Process Control (SPC) framework for high-dimensional streaming data.
 - **1.4 Summary of Contributions:**
 - Mathematical proof of chunk excursion bounds (Theorems 1–2, Corollary 1) and multivariate FWER control (Theorem 3).
 - $O(D)$ Bounded RAM Architecture (**3.23 KB**, $180\times$ savings).
 - 4-Stage Statistical Pipeline (Spike Filter, Lazy Expansion, MLE Candidate Fitting, Bonferroni Adjustment).
 - Unified Two-Tier Benchmark (10 1D distributions, 4 economic datasets, 5 industrial streams, TEP Modes 1, 3, 4, 5, and Sensitivity Analysis).
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Section 2: Related Work & Literature Review

- **2.1 Classical & Multivariate SPC Control Charts:** Shewhart X-Bar, EWMA, CUSUM, Hotelling's T^2 — mathematical breakdown under non-normality and non-stationary noise.
 - **2.2 Bootstrap Resampling in Streaming Environments:** Efron's percentile bootstrap, blockwise bootstrap, online bootstrap for SGD — high memory footprint and vulnerability to buffer contamination.
 - **2.3 Uncertainty Quantification & Outlier Filtering in IoT Data Streams:** Review of range estimation, hyper-ellipsoid bounds, and z-score spike filtering.
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Section 3: Theoretical Foundations and Univariate Streaming Range Estimation (Part I - From Draft 1 PDF)

- **3.1 Streaming Data Model & Discard-After-Learn Constraint:** Chunks $c_i \in \mathbb{R}^{m \times D}$ arriving over time $t = 1, 2, \dots$
- **3.2 Univariate Population Range Estimation ($[L, R]$):** Formulation of left-end (L) and right-end (R) boundary expansion.
- **3.3 Mathematical Analysis of Extreme Value Excursion:**
 - **Theorem 1 (First Chunk Extreme Probability):** Proof of joint minimum/maximum occurrence probability in initial chunk:

$$P(E_1) = \frac{l(l-1)}{N(N-1)}$$

- **Theorem 2 (Streaming Chunk Distribution):** Proof of uniform extreme occurrence probability across streaming chunks for $i \geq 1$:

$$P(E_i) = \frac{l(l-1)}{N(N-1)}$$

- **Corollary 1 (Expected Chunk Index):** Derivation of expected chunk index $\mathbb{E}[X]$ for initial boundary expansion:

$$\mathbb{E}[X] = \frac{1}{2} \left(\frac{l(l-1)L(L+1)}{N(N-1)} \right)$$

Section 4: Proposed High-Dimensional RBULT-SPC Framework (Part II - From research_plan.md)

- **4.1 System Architecture:** 4-Stage streaming pipeline.
- **4.2 Algorithm 1: First Chunk Initial Fitting & Tail Bin Partitioning**
- **4.2.1 Akaike Information Criterion (AIC) Selection & Streaming Efficiency Analysis:**
 - **Mathematical Definition & Formula:**

$$\text{AIC} = 2k - 2 \ln(\hat{L})$$

where k represents the number of estimated parameters (parsimony penalty) and $\hat{L}$ represents the maximum likelihood of the candidate distribution. Minimizing AIC selects the optimal non-parametric distribution that maximizes goodness-of-fit while preventing overfitting.

- **Phase 1 Initial Fitting Role (c_1):** Evaluates Maximum Likelihood Estimation (MLE) and AIC across 11 continuous candidate distributions $\mathcal{P} = \{\text{Gamma, Weibull, Wald, Uniform, Rayleigh, Normal, } \dots\}$ on the initial data chunk c_1 . The candidate distribution $f(x; \hat{\theta}) \in \mathcal{P}$ minimizing AIC ($\arg \min \text{AIC}$) is selected as the baseline template to compute theoretical tail element counts $|h_1|$ and $|h_8|$ via exact CDF integration.
- **Phase 2 Streaming Recurrence Efficiency ($c_m, m > 1$):** AIC model selection is **NOT** re-evaluated for every streaming chunk c_m in normal in-control operations. Re-evaluating MLE and AIC across 11 candidate distributions per chunk would introduce prohibitive $O(N_{\text{models}} \cdot |c_m|)$ computational latency, violating real-time streaming constraints. Instead, RBULT reuses the established standard histogram template H and relies on **Algorithm 3 (Bootstrap Tail Resampling)** and **Algorithm 4 (Z-Score Outlier Filter)** for fast lazy boundary adjustments (4.18 – 4.78 ms per chunk).
- **Concept Drift Refitting Trigger (Paper 2 Roadmap):** AIC model selection is re-triggered **only** when an external concept drift detector (e.g., ADWIN / DDM) signals a permanent distributional shift in the data stream, updating the baseline template H .
- **4.3 Algorithm 2: Streaming Chunk Update & Lazy Boundary Expansion**
- **4.4 Algorithm 3: Recurrent Tail Resampling & Quantile Interpolation**
- **4.5 Algorithm 4: Z-Score Outlier Spike Filtering:** Suppression of contaminated sensor noise to prevent control limit pollution.
- **4.6 Bonferroni / Šidák FWER Multi-Dimensional Scaling:** Tail boundary adjustment $\alpha_{\text{dim}} = \frac{\alpha_{\text{sys}}}{D}$ for D -dimensional feature spaces.
- **4.6.1 Mathematical Evolution of Control Limits (1D Range vs. Dimension-Aware LCL/UCL):**

Formula Comparison: Draft 1 (1D Range) vs. Draft 2 (Multivariate RBULT-SPC)

Mathematical Dimension	Draft 1 (Univariate 1D Baseline in PDF)	Draft 2 (Multivariate RBULT-SPC in Second Draft)
Target Data Dimension	Single Scalar Feature ($D = 1$)	Multivariate Sensor Stream ($D \geq 1$, e.g., $D = 34$)
Tail Quantile Adjustment	Univariate Tail Probability α ($D = 1 \Rightarrow \frac{\alpha}{1} = \alpha$)	Bonferroni FWER Scaled Quantile: $\alpha_{\text{dim}} = \frac{\alpha_{\text{sys}}}{D}$
Lower Control Limit (LCL _d)	$L = \text{Percentile}_{\text{left}}\left(\frac{\alpha}{2}\right)$	$\text{LCL}_d = \mathbf{L}_d = \mathbf{Q}_{\text{tail}}\left(\frac{\alpha_{\text{sys}}}{2D}\right)$

Mathematical Dimension	Draft 1 (Univariate 1D Baseline in PDF)	Draft 2 (Multivariate RBULT-SPC in Second Draft)
Upper Control Limit (UCL_d)	$R = \text{Percentile}_{\text{right}}\left(1 - \frac{\alpha}{2}\right)$	$UCL_d = R_d = Q_{\text{tail}}\left(1 - \frac{\alpha_{\text{sys}}}{2D}\right)$
Outlier & Noise Defense	Basic z-score outlier elimination	Algorithm 4 Z-Score Filter integrated prior to Q_{tail} calculation

💡 Key Scientific Rationale for the Mathematical Extension:

- Univariate ($D = 1$) vs. Multivariate ($D = 34$) Scaling:** In Draft 1, all 10 simulated distributions and 4 Kaggle datasets were 1-dimensional ($D = 1$), so the tail quantile scaling factor was $\frac{\alpha}{1} = \alpha$. In Draft 2, expanding to $D = 34$ sensor channels requires dividing by D ($\alpha_{\text{dim}} = \frac{\alpha_{\text{sys}}}{D}$) to constrain system-wide false alarms to $\leq \alpha_{\text{sys}} = 5\%$, preventing an explosion of false alarm spam ($1 - (1 - 0.05)^{34} \approx 82.5\%$).
 - Eliminating Control Limit Pollution:** Integrating Algorithm 4 Z-score filtering before computing Q_{tail} ensures that transient sensor spikes or long-duration faults do not distort L_d and R_d .
- 4.7 Theorem 3 (Family-Wise Error Rate Bound):** Formal proof of FWER upper bound control under Bonferroni tail adjustment in D -dimensional non-Gaussian space.
 - 4.8 Complexity Analysis:**
 - Space Complexity: Strictly $O(D)$ bounded RAM (3.23 KB).
 - Time Complexity: Amortized $O(1)$ per sample (< 45 ms per chunk).

Section 5: Two-Tier Empirical Benchmark Experiments

- 5.1 Experimental Setup & Evaluation Metrics:** Range Error, Coverage Rate (%), Sample FAR (%), Chunk FAR (%), ARL0, ARL1 (Detection Delay), Peak RAM (KB), Latency per Chunk (ms).
- 5.2 Tier 1: Univariate Range Approximation & Dual-Protocol Noise Sensitivity Benchmarks:**
 - Evaluates 1,260 simulation runs across 5 synthetic distributions ($F(5, 10)$, Uniform, Wald, Gamma, Normal), 3 chunk sizes (50, 100, 500), 2 target alphas ($\alpha = 0.05, 0.01$), 3 methods, and 2 protocols.
 - Dual Evaluation Protocol:**
 - In-Sample Adaptation Protocol:* Measures post-update boundary enclosure precision.

2. One-Step-Ahead Pre-Sequential Protocol (IEEE TKDE Gold Standard):

Measures predictive violation rates using bounds from chunk $m - 1$ prior to updating chunk m .

• 7-Scenario Gold Standard Suite:

- *Scenario A (Clean Stream)*: Baseline pure stream (Coverage = 95.41%, $\bar{W} = 23.65$, Latency = 4.45 ms).
- *Scenarios B1–B3 (GAWN $0.1\sigma, 0.2\sigma, 0.3\sigma$)*: Scale-invariant continuous noise adaptation (Coverage = 95.34% – 95.74%, NSR = 1.05 – 1.21).
- *Scenario C1 (1% Spikes)*: Perfect transient glitch absorption (Coverage = 95.29%, $\bar{W} = 24.22$, $\sigma_L = 0.05$).
- *Scenarios C2–C3 (5% & 10% Spikes)*: Extreme stress resilience preventing memory explosion and boundary collapse (Coverage = 97.70% – 99.44%, $\sigma_L = 0.34$).

- 4 Real-World Economic Datasets (Laptop Prices, Electronic Sales, E-Commerce Sales, World Tourism Economy).

• 5.3 Tier 2: Real-World Industrial Streaming Benchmarks (From `research_plan.md`):

- AI4I 2020 Predictive Maintenance ($N = 10,000$, $D = 5$)
- MetroPT-3 Air Compressor ($N = 1,516,948$, $D = 7$)
- Large Industrial Pump Maintenance ($N = 20,000$, $D = 5$)
- Water Pump Sensor Dataset ($N = 220,320$, $D = 10$)

• 5.4 Tier 2 Stress Benchmark: Tennessee Eastman Process (TEP Multi-Mode Matrix):

- Mode 1: Nominal Operating Conditions (50/50 Mass Ratio, Nominal Throughput)
- Mode 3: Chemical Feed Skewness (90/10 Mass Ratio)
- Mode 4: Operational Throughput Stress (50/50 Mass Ratio, Max Production Rate)
- Mode 5: Combined Extreme Stress (10/90 Mass Ratio, Max Production Rate)

• 5.5 Hyperparameter Sensitivity Study (`ooc_threshold_count` $\in \{5, 10, 15\}$):

Demonstration of zero false alarm spam (Chunk FAR = 0.00%) and immediate detection ($ARL_1 = 1.00$).

- **5.6 Resource & Latency Trade-off Analysis:** $180\times$ RAM savings vs. < 45 ms real-time execution.

Section 6: Discussion & Data Taxonomy Insights

- **6.1 Data Taxonomy Synthesis:** Performance across Skewed/Heavy-Tailed, Uniform Density, High Throughput Stress, and High Dimensionality ($D = 34$).

- **6.2 Methodological Synthesis:** Comparing Tier 1 range error accuracy with Tier 2 multivariate SPC control chart coverage.
- **6.3 Practical Engineering Guidelines for Smart Factory IoT Implementations.**

Section 7: Conclusion & Future Scope

- **7.1 Summary of Scientific Breakthroughs.**
- **7.2 Roadmap for Paper 2:** Integrating Concept Drift Detection (ADWIN/DDM) and Dynamic Model Refitting for long-term non-stationary streams.

 Summary Table of Data Coverage in Unified Manuscript

Tier / Level	Dataset Name	Sample Size (<i>N</i>)	Dimensions (<i>D</i>)	Data Type & Distribution	Source
Tier 1	1D Benchmark (7-Scenario Suite)	1,260 Runs (<i>N</i> = 10,000)	1	<i>F</i> , Uniform, Wald, Gamma, Normal	Section 10 & PDF
Tier 1	4 Economic Kaggle Datasets	1,000 - 10,000	1	Laptop, Electronic, E-Com, Tourism	Draft 1 PDF
Tier 2	AI4I 2020 Predictive Maint.	10,000	5	Mechanical Tool Wear Sensor	research_plan.mc
Tier 2	MetroPT-3 Air Compressor	1,516,948	7	Heavy-Tailed Pneumatic Stream	research_plan.mc
Tier 2	Large Industrial Pump	20,000	5	Uniform Density Maintenance	research_plan.mc
Tier 2	Water Pump (sensor.csv)	220,320	10	Multi-Sensor Industrial Stream	research_plan.mc

Tier / Level	Dataset Name	Sample Size (N)	Dimensions (D)	Data Type & Distribution	Source
Tier 2	TEP Mode 1 (Nominal)	1,740,000	34	Chemical Plant Nominal Stream	research_plan.mc
Tier 2	TEP Mode 3 (Feed Skew 90/10)	1,739,400	34	Chemical Feed Skewness Stream	research_plan.mc
Tier 2	TEP Mode 4 (Max Production Rate)	1,719,000	34	Operational Throughput Stress	research_plan.mc
Tier 2	TEP Mode 5 (Combined Stress)	1,729,800	34	Combined Extreme Stress Stream	research_plan.mc

💡 Additional Strategic Comments & Reviewer Framing Notes

🎯 1. Rationale for Two-Tier Experimental Hierarchy (Tier 1 vs. Tier 2)

Structuring the experimental evaluation into **Tier 1** (Univariate Controlled Benchmarks) and **Tier 2** (Multivariate Streaming SPC Benchmarks) is a standard high-impact strategy for top Q1 journals (*Information Sciences*, *IEEE TKDE*):

- **Satisfying Theoretical Reviewers (Tier 1):** Validates the fundamental RBULT tail-fitting engine ($[L, R]$ range estimation accuracy) under 10 controlled 1D synthetic distributions and 4 economic datasets where mathematical ground-truth bounds are known.
- **Satisfying Applied/Systems Reviewers (Tier 2):** Validates the operational streaming process control performance (Coverage %, Sample FAR %, Chunk FAR %, ARL1, Memory KB, and Latency ms) on 5 real industrial telemetry streams and 4 TEP stress regimes ($D = 34$, up to 1.74M observations).

- **Preventing Common Reviewer Objections:** Eliminates common rejection risks such as "lack of ground-truth controlled synthetic validation" or "lack of high-dimensional real-time streaming applicability".
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2. Recommended Framing Passage for Section 5.1 (Experimental Setup)

Insert this bridging text at the start of Section 5.1 to guide reviewers seamlessly from Tier 1 to Tier 2:

"To rigorously evaluate the proposed RBULT-SPC framework from both fundamental estimation accuracy and operational streaming process control perspectives, our empirical study is structured into a progressive two-tier experimental hierarchy:

- **Tier 1 (Base Estimation Accuracy Benchmark):** Evaluates the fundamental range estimation accuracy of the RBULT engine ($pop_range - est_range$) across 10 controlled 1D synthetic distributions and 4 economic datasets where theoretical ground-truth bounds are known.
 - **Tier 2 (Multivariate Streaming SPC Benchmark):** Evaluates operational process control metrics (Coverage Rate %, Sample FAR %, Chunk FAR %, ARL1, Peak RAM, and Latency) across 5 real-world industrial telemetry streams and 4 Tennessee Eastman Process stress regimes ($D = 34$, up to 1.74M observations)."
-

3. Mathematical Parameter Evolution Note (α & Bonferroni FWER Scaling)

- **Draft 1 Baseline ($D = 1$):** Evaluated single scalar features where the tail percentile parameter was unscaled ($\alpha_{\text{dim}} = \frac{\alpha}{1} = \alpha$).
- **Draft 2 Extension ($D \geq 1$):** Extends the theoretical formulation to multivariate streaming control limits by introducing Bonferroni FWER tail scaling ($\alpha_{\text{dim}} = \frac{\alpha_{\text{sys}}}{D}$). This constrains system-wide false alarms to $\leq \alpha_{\text{sys}} = 5\%$, preventing an explosion of false alarm spam ($1 - (1 - 0.05)^{34} \approx 82.5\%$).